METRIC

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COMMERCIAL ITEM DESCRIPTION

DRY FILTER UNIT

The General Services Administration has authorized the use of this commercial item description, for all federal agencies.

1. SCOPE.

1.1 Scope.

This commercial item description covers a Dry Filter Unit (DFU). The DFU is an ambient air sampler that can be used to collect airborne biological particles. Those particles may be transferred to a liquid medium, and analyzed for the presence of possible biological warfare agents. The DFU is available in two types or configurations. The Type I DFU (also known as DFU-1000) is intended for use in most environmental conditions. The Type II DFU (also known as DFU-2000) is intended for use in nearly all environments. It is better suited for operation in special environments in which air is laden with precipitation, sand, or debris. Both DFU sampler types may operate indoors or outdoors in military and non-military applications.

AMSC N/A FSC 6665

Beneficial comments, recommendations, additions, deletions, clarifications, etc. and any other data which may improve this document should be sent to: U.S. Army Edgewood Chemical Biological Center, ATTN: AMSRD-ECB-ENA-S, E3331, 5183 Blackhawk Road, Aberdeen Proving Ground, MD 21010-5424 or emailed to SpecsTeam@apgea.army.mil.

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2. CLASSIFICATION.

2.2.1 Type I DFU. Type I DFU includes a blower which draws air from above the unit through a manifold which contains removable filter cartridges. Filtered air discharged from the blower is exhausted through an outlet near the bottom of the DFU. The blower, power cord, spare parts, and consumables are contained within an impact-resistant, portable case equipped with hand-turn fasteners for opening and closing.

2.2.2 Type II DFU. Type II DFU includes the same components of the Type I DFU. All Type I components are contained within a larger, weather-resistant, secondary enclosure. A stack is mounted on top of the enclosure. A special air inlet pre-separator or shroud on top of the stack is intended to prevent the entrance of precipitation, sand, and debris (e.g. leaves, insects).

3. SALIENT CHARACTERISTICS.

3.1 Operating characteristics.

Unless otherwise specified, the following requirements apply to both types of DFUs:

3.1.1 General operation.

The DFU shall operate by passing ambient air through filters and collecting airborne particles, manual transfer of particle-laden filters to a conical tube containing phosphate buffer solution, manually shaking or centrifuging in a conical tube, and manual transfer of liquid droplets to a biological identifier (hand held assay panel). The DFU shall be capable of operating indoors and outdoors on land and aboard ships.

3.1.1 Flow rate.

a. Combined air flow through two filters installed in parallel in the intake manifold of the DFU shall be at least 900 liters per minute (lpm).
b. Air flow through the DFU with no filters installed in the intake manifold shall be at least 1250 lpm at standard conditions or corrected to standard conditions. In this document, standard conditions are defined as one atmosphere (sea level conditions), and a temperature of 20°C (68°F).

3.1.2 Filter efficiency.

a. During operation, each filter shall be able to trap at least 97% of passing particles whose size is one micron or greater.
b. During operation at rated flow, the filter shall not experience tears or holes when dry or wet.

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3.1.3 Extraction efficiency.

The extraction or elution process, whereby the particles containing biological material collected by the filter are transferred into the Phosphate Buffer Solution (PBS), shall have an extraction efficiency of at least 50%.

3.1.4 Continuous operation.

The DFU shall be capable of continuously collecting material from air samples for at least 8 hours before replacing the collection filter(s).

3.1.5 Power consumption.

The DFU shall be capable of operating using 120 or 240 Volts Alternating Current (VAC) with frequency of 50 or 60 Hz supplied by commercial line power or generator input power. CONUS units are shipped with 120 VAC power cord only while OCONUS units are shipped with both 120 and 220 VAC power cords. When input voltage is 120 VAC, 60 Hz, current shall not exceed 5.0 amperes (except peak power draw shall not exceed 17 amperes during startup).

3.1.6 Noise levels.

DFU operating noise levels shall not exceed 60 dB at 61 cm (2 feet) when measured with the DFU in an open-air environment void of any additional ambient noise.

3.1.7 Weight.

The Type I and II DFU shall be man portable. The maximum weight of the Type I DFU shall be 19.1 kilograms (42 pounds) and the maximum weight of the Type II DFU shall be 38.1 kilograms (84 pounds) without accessories (pre-separator shroud, stack assembly, power cord and its bag, filter cartridges). Both versions shall be capable of being operated and maintained by personnel representative of the 5th and 95th percentile.

3.2 Interface and interoperability characteristics.

This section includes a brief description of both external and internal interfaces. It also includes some component functional characteristics in relation to their surrounding parts.

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3.2.1 External power interface connections.

Plug at end of cable shall be rated at 15 amps or above. Plug connector details shall be in accordance with Table I and MIL-DTL-5015 for any input AC voltage.

TABLE I. Power cable/AC input interface connector and pin-out.

Connector Description	Reference Designator	Pin-Out	
Connector 3-position	AC Input	3	Chassis GND
		2	NEUT
		1	Applicable* VAC

* The DFU shall be capable of operating using 120 or 240 Volts Alternating Current (VAC) with frequency of 50 or 60 Hz supplied by commercial line power or generator input power (CONUS units shipped with 120 VAC power cord only while OCONUS units shipped with both 120 and 220 VAC power cords).

3.2.2 Internal power interface connections.

This interface allows power input and control from the power source to the DFU. Table II defines the interface between the power cable and DFU. The DFU's connector interfaces with the power cable's AC output connector. The DFU's connector has male threads and 6 pins. The cable's interfacing connector has female threads and 6 sockets. Connectors shall have an Ingress Protection (IP) number of 68 (dust tight and protected against immersion). Connector enclosures shall conform to rating 6P in NEMA Standard Publication 250 and IEC 60529. The DFU's power inlet connector will have a captive cover to be used when DFU is not in operation.

TABLE II. DFU power/ interface connector and pin-out.

External Connector	Reference Designator	Connector Description	MIL-SPEC		Mating MIL-SPEC	
DFU		Connector, 6-position	MIL-DTL-38999		MIL-DTL-38999	
Pin #	**Signal**			**I/O**	**Signal Bus**	
1	120 VAC			I	220 VAC POWER	
2	NEUT			I	120 VAC POWER	
3	GND			I	GND	
4	240 VAC			I	NC	
5	NEUT			I	Neutral	
6	GND			I	NC	

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3.2.3 Manifold, air intake.

a. Air sampled from external environment is drawn through the air intake manifold by negative pressure and induced airflow generated by the blower. The air intake manifold shall be located on the top of the Type I DFU for the most representative sample. The manifold shall provide an air routing path through 2 filter cartridges installed by the operator within the manifold.

b. The manifold shall mate to a fitting on case (enclosure) to form a sealed air path. The manifold shall mount vertically to a “female” bayonet-type fitting in the top of the DFU case. The manifold shall feature a “male” bayonet-type fitting to lock the manifold in place by pushing down 1.9 cm (¾ inch) and rotating 25 degrees in either direction.

c. When DFU is not is use, a captive cover shall be capable of sealing intake of manifold and preventing debris from entering manifold.

3.2.4 Filter interface.

a. Filter diameter shall be 47.0 mm (1.85 inch) for interface with filter cartridge. Deviations may be allowed if it is demonstrated that no air circumvents filter.
b. Each filter shall be easy to insert, remove, and secure in place using filter cartridge.

3.2.5 Filter cartridge.

a. DFU kit shall contain 4 filter cartridges. Two are installed during operation.
b. Construction material shall be resistant to particle accumulation and cleaning fluids.
c. Each filter cartridge shall be capable of holding a single filter membrane or rubber stopper.
d. Each filter cartridge shall slide horizontally into side of manifold prior to use.
e. Each filter cartridge shall slide horizontally outward and be packed in the DFU case while not in use.

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3.2.6 DFU consumables.

Table III gives a brief description of each of the mission consumables. For spare part description, see 3.4.4.

TABLE III. DFU mission consumables

Consumable	Description	Function and interface
Conical Tube of PBS	Contains Phosphate Buffer Solution (10 ml minimum)	To transfer bio material particles to liquid medium. To allow mechanical agitation.
Filter Pack	Contains 3 filters	To provide replacement filters for use during dry sample collection.
Dry Filter Unit Sampling Kit	Each kit contains pipet, latex gloves, whirl bags, and filter pack	To provide materials necessary for dry sample collection and conversion to wet sample for analysis.
Hand Held Assay (HHA)	A device containing numerous inoculation strips which may be wetted and optically scanned	To identify the type of biological material present.

3.2.7 Internal interface characteristics for Type II DFU.

The Type II DFU shall use the basic components of the Type I DFU. Type I DFU components shall be contained within a larger, weather-resistant, secondary enclosure whose size is 55.9 cm wide by 42.2 cm deep by 81.3 cm high, plus or minus 0.1 cm, (22.0” wide by 16.6” deep by 32.0” high, plus or minus 0.04”) without pre-separator stack. A stack shall be mounted on top of the enclosure. A special air inlet pre-separator with shroud on top of the stack is intended to prevent the entrance of precipitation, sand, and debris (e.g. leaves, insects). Total height of Type II DFU with stack and shroud shall be 2.7 meters (9.0 feet).

3.3 Environmental characteristics.

The DFU (Type I and II) shall operate and deploy without degradation under the following climatic conditions:

a. Operating temperatures of -28 to 60°C (-20 to 140°F).
b. Storage temperatures of -39 to 62°C, indefinite time.
c. Altitudes of up to 2.44 km (8000 ft) above sea level.
d. Solar radiation.
e. 5 to 95% relative humidity.

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The Type II DFU shall be resistant to extreme external conditions such as wind, rain, snow, corrosion, sand and large dust particles. It must prevent such naturally occurring substances and other debris from restricting filter airflow.

3.4 Support and ownership characteristics.

3.4.1 Security.

The Type II DFU shall be able to be secured, preventing any unauthorized movement or entry into the unit. Additionally, the filters or samples collected shall be protected from removal, tampering, or access by unauthorized personnel.

3.4.2 Safety.

a. The DFU sampler shall be safe for transport, storage, assembly, operation, and disposal throughout its life cycle. It shall be capable of unrestricted air, highway, rail and marine transport worldwide.

b. The DFU shall be certified by UL and shall have a UL approved marking. Consider electrical, shock and fire hazards and degree of protection afforded by enclosures in various environments.

3.4.3 Reliability.

Time to failure of DFU shall not be less than 2 years of calendar time or 14016 hours of continuous operation (assuming 24 operating hours per day and 80% utilization). Consumable replacement is not counted as a failure.

3.4.4 Maintainability.

Parts identified as spare parts shall be easily replaced. Spare parts need not be identical but shall be interchangeable in terms of form, fit and function. Performance and reliability of any spare part shall be at least as high as the one being replaced. The following parts of the DFU kit are identified as spare parts and must be available:

Type I:
- Intake manifold
- Filter pack
- Filter cartridge
- Gasket/o-ring kit
- Cable (120 VAC)
- Cable (220 VAC)
- Dry filter unit sampling kit

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Type II:
- Outer enclosure
- Air inlet pre-separator and stack
- Intake manifold
- Filter pack
- Filter cartridge
- Gasket/o-ring kit
- Cable (120 VAC)
- Cable (220 VAC)
- Dry filter unit sampling kit

3.4.5 Markings.

3.4.5.1 Hazard warnings. If any use conditions create a health hazard potential, the equipment shall have appropriate hazard warning labels in accordance with 29 CFR 1910.1200.

3.4.5.2 Identification markings. Markings on the outside surface of DFU case shall include the name (Dry Filter Unit), model number, contract number, lot number, serial number, cage code, and manufacturer's name. Suggested position of labels is shown in Figure 1 and 2.



FIGURE 1. Type I DFU with labels

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FIGURE 2. Type 2 DFU with labels

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4. REGULATORY REQUIREMENTS.

The offeror/contractor is encouraged to use contaminant-free materials to the maximum extent practical. Materials at the time of disposal shall not be a Resource Conservation Recovery Act (RCRA) hazardous waste as defined in 40 CFR 261.21 - 261.24 or 40 CFR 261.33 (e) and (f).

5. PRODUCT CONFORMANCE.

5.1 Product conformance.

The products provided shall meet the salient characteristics of this commercial item description, conform to the producer's own drawings, specifications, standards, and quality assurance practices, and be the same product offered for sale in the commercial market. The Government reserves the right to require proof of such conformance.

5.2 Market acceptability.

The DFU offered must have been sold to the government or commercial market for at least 12 months.

5.3 Product demonstration.

Unless otherwise specified in the contract or order (see 7.1), an 8-hour first article operating demonstration shall be conducted on 3 DFUs. Each DFU should operate continuously at least 4 hours at each nominal input voltage (120 and 240 VAC).

6. PACKAGING.

Preservation, packing, labeling, and marking shall be as specified in the contract or order.

7. NOTES.

7.1 Ordering data.

Acquisition documents should specify the following:

a. Title, number, and date of this commercial item description.
b. DFU configuration (Type I or II).
c. Type of input power and length of power cable.
d. Any special air inlet or air exhaust interface adapters between DFU and host, if applicable.
e. Packaging requirements.
f. Whether a first article inspection on product samples or a product demonstration is required (see 5.3). When military packaging is required by the contract, first article inspection should be required.

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7.2 Technical data.

Other technical data is contained Technical Manual No. 3-6665-348-12&P available from Joint Product Manager, Biological Detection, ATTN: SFAE-BD, Aberdeen Proving Ground, MD 21010-5424.

Blower should be Model No. VFC300P-5T manufactured by Fuji Electric or equivalent.

Filter media should be 1 micron polyester felt.

7.3 Sources of documents.

7.3.1 Federal standards.

a. MIL-DTL-5015 – Connectors, Electrical, Circular Threaded, AN Type, General Specification for
b. MIL-DTL-38999 – Connectors, Electrical, Circular, Miniature, High Density, Quick Disconnect, Environment Resistant, Removable Crimp and Hermetic Solder Contacts, General Specification for

Copies of these documents are available online at http://assist.daps.mil/quicksearch/ or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.

7.3.2 Code of Federal Regulations.

40 CFR Part 261 - Identification and Listing of Hazardous Waste
29 CFR Part 1910.1200 – Hazard Warnings

Copies of Code of Federal Regulations are available on-line at http://www.access.gpo.gov.or from the U.S. Government Printing Office, Washington, DC 20402.

7.3.3 Non-Government documents.

NEMA - Standards Publication 250, Enclosures for Electrical Equipment
IEC 60529 – Degree of Protection Provided by Enclosures

Copies of these documents are available on-line from National Electrical Manufacturers Association at http://www.nema.org/stds/ and from International Electrotechnical Commission (IEC) at http://www.techstreet.com/info/iec.tmpl .

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7.4 Keywords.

Airborne particulate
Detection
Identification
Biological warfare agents

MILITARY INTERESTS:

Custodian:
Army - EA
Navy – SH
Air Force - 99

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA - FSS
VA - OSS

Review activity:

DLA - GS

Preparing activity:

Army - EA
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